

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA07 **COURSE CODE:** Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

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Code of Conduct:

I ___S>HARIPRIYA_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

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Signature of the student:

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→ DataLinkLayer → ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

Question 1 — Missing Layers in the OSI Concept Map

Given map: *Application* → *Presentation* → ____ → *Transport* → *Network* → ____ → *Physical*.

Concept Map

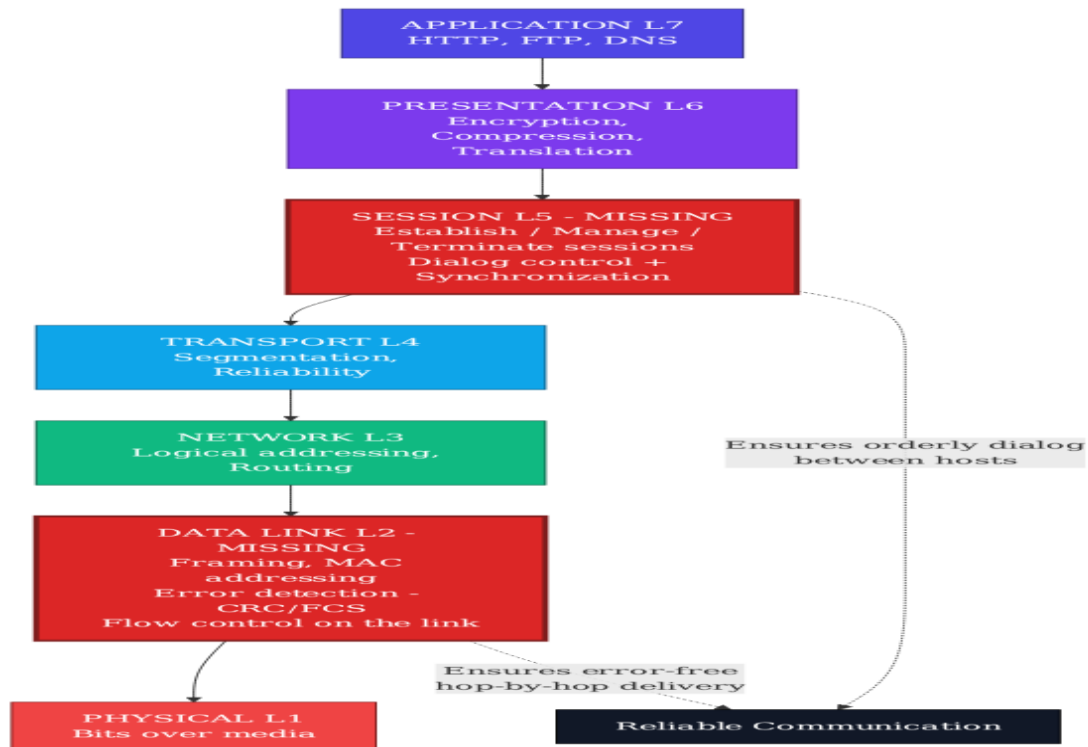


Figure 1 — Missing OSI layers (Session & Data Link) shown in red.

Missing Layers Identified

- Between Presentation and Transport → SESSION LAYER (L5)
- Between Network and Physical → DATA LINK LAYER (L2)

Functions Essential for Reliable Communication

Session Layer (L5)

- Establishes, manages and terminates sessions between applications.
- Provides dialog control (half/full duplex) and synchronization checkpoints so long transfers can resume after failure.
- Without it, applications cannot coordinate multiple exchanges reliably.

Data Link Layer (L2)

- Frames raw bits and adds MAC addresses for hop-by-hop delivery.
- Performs error detection using CRC/FCS and provides flow control on the link.
- Without it, bits from the Physical layer cannot be interpreted, checked, or delivered to the correct local device.

Together, the two missing layers bridge application-level dialog and physical bit transmission, making end-to-end reliable communication possible.

Question 2 — Data Units and OSI Layers

Concept Map

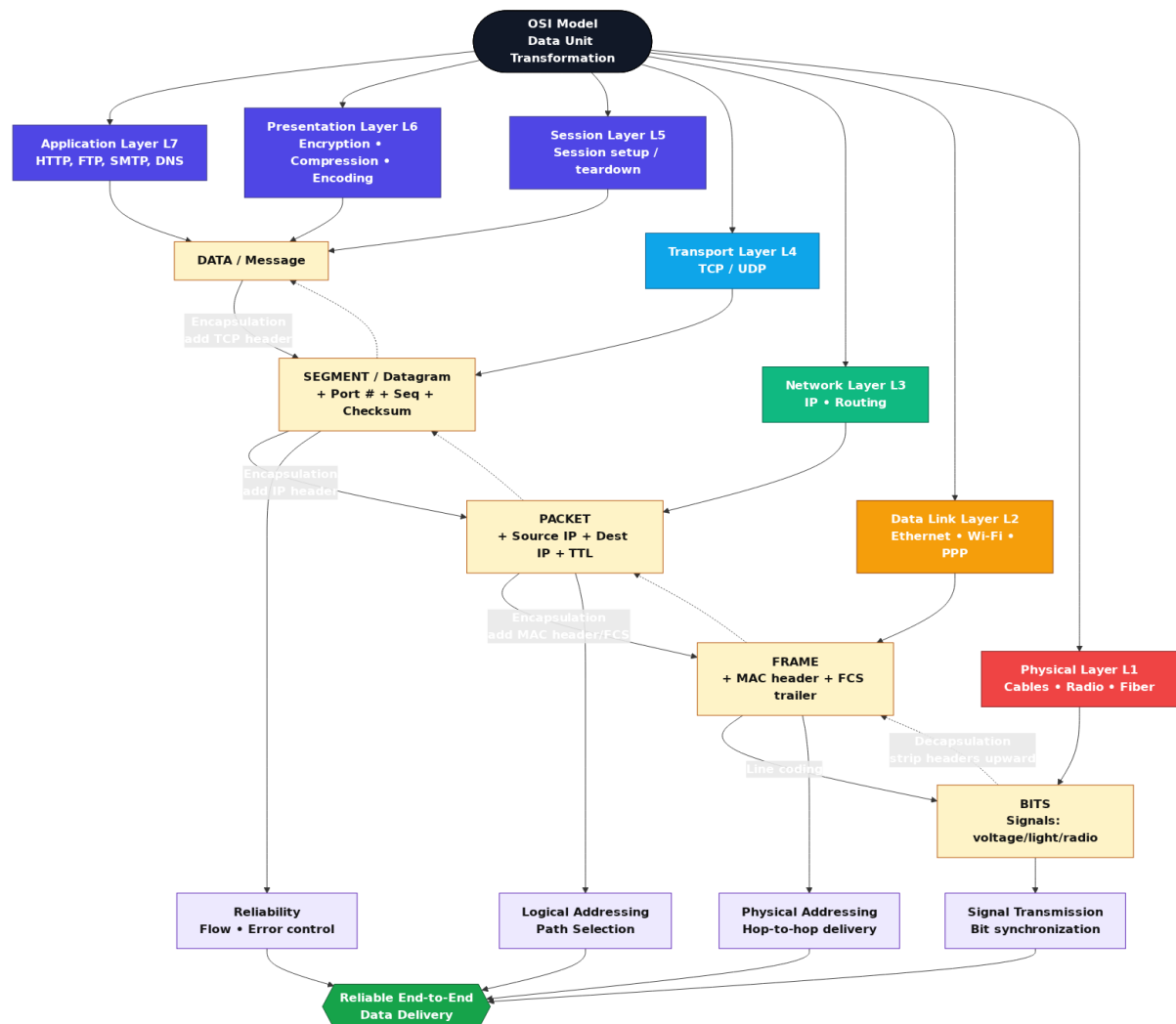


Figure 2 — Encapsulation: Data → Segment → Packet → Frame → Bits.

Data Units mapped to OSI Layers

Data Unit (PDU)	OSI Layer & Role
Data / Message	Application, Presentation, Session (L7–L5) — user information
Segment / Datagram	Transport (L4) — adds port numbers, sequence numbers, checksum
Packet	Network (L3) — adds source & destination IP for routing
Frame	Data Link (L2) — adds MAC header + FCS trailer for local delivery

Bits	Physical (L1) — encoded as electrical/optical/radio signals
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Analysis — How Transformation Supports Transmission

- Encapsulation adds control information at every layer, so each layer at the receiver can independently do its job.
- Segmentation at L4 breaks large data into manageable pieces and enables reliability (ACK, retransmission).
- Packetization at L3 makes routing across different networks possible using logical IP addresses.
- Framing at L2 enables error detection and correct local delivery using MAC addresses.
- Bit-level encoding at L1 allows the message to physically travel across media.
- At the receiver, decapsulation reverses the process (Bits → Frame → Packet → Segment → Data) delivering original data intact.

Question 3 — MAC & IP: Local vs Global Delivery

Concept Map

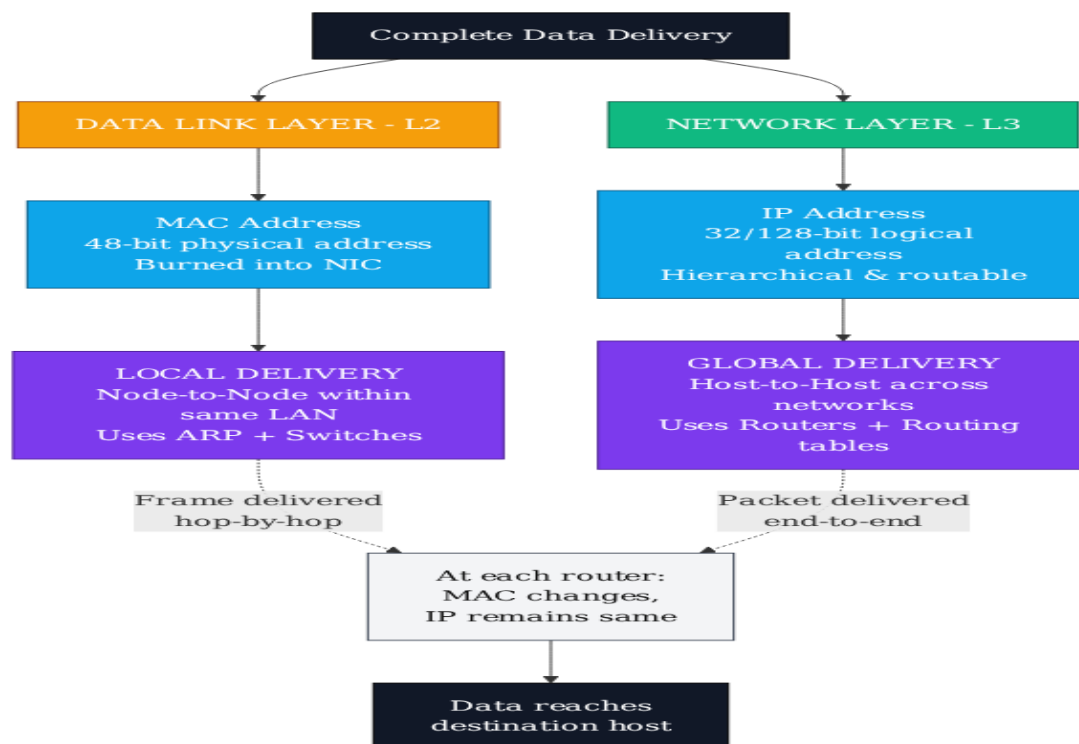


Figure 3 — Data Link/MAC handles local delivery; Network/IP handles global delivery.

Relationships Explained

Aspect	Details
Data Link → MAC → Local Delivery	MAC (48-bit physical address) delivers a frame to the correct device on the same LAN using switches and ARP.
Network → IP → Global Delivery	IP (logical, hierarchical address) delivers a packet across multiple networks using routers and routing tables.
Interaction at each hop	IP address stays the same end-to-end; MAC address changes at every router hop as the frame is rebuilt for the next link.

Analysis

- MAC addressing alone cannot reach a device on another network — it has no concept of routing.
- IP addressing alone cannot deliver a packet on the final wire — routers need MAC to hand off frames locally.
- Together, IP identifies WHERE (which network/host) and MAC identifies HOW (which local NIC) — ensuring complete end-to-end data delivery.

Question 4 — Error Control across Data Link & Transport Layers

Concept Map

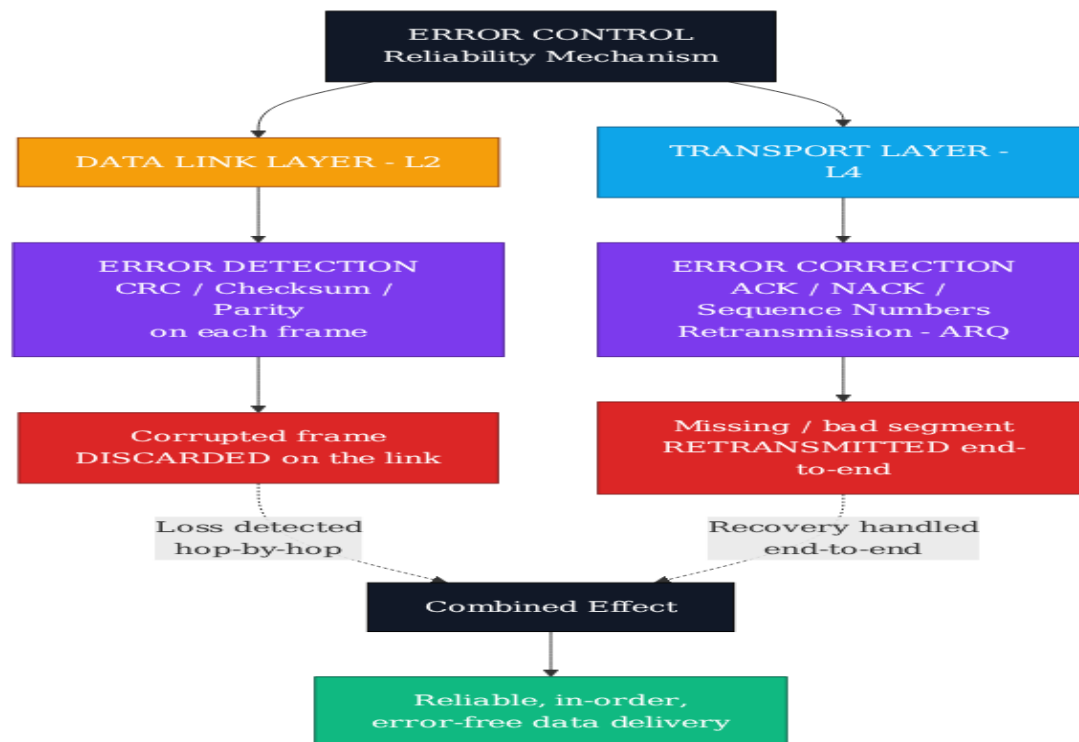


Figure 4 — L2 detects errors hop-by-hop; L4 corrects errors end-to-end.

Layer Responsibilities

Layer	Error Control Role
Data Link (L2)	Error DETECTION using CRC / checksum / parity on each frame. Corrupted frames are discarded on the link.
Transport (L4)	Error CORRECTION using acknowledgements, sequence numbers, and retransmission (ARQ, e.g. TCP).

Analysis — Combined Reliability

- L2 catches bit-level errors early, on every physical hop, preventing corrupted frames from wasting upstream resources.
- L4 handles losses that L2 cannot see — dropped packets, out-of-order arrivals, congestion — end-to-end between source and destination.
- Their interaction produces layered reliability: local integrity (L2) + global recovery (L4) = error-free, in-order delivery to the application.

Question 5 — Web Browsing Concept Map & Layer Failures

Concept Map

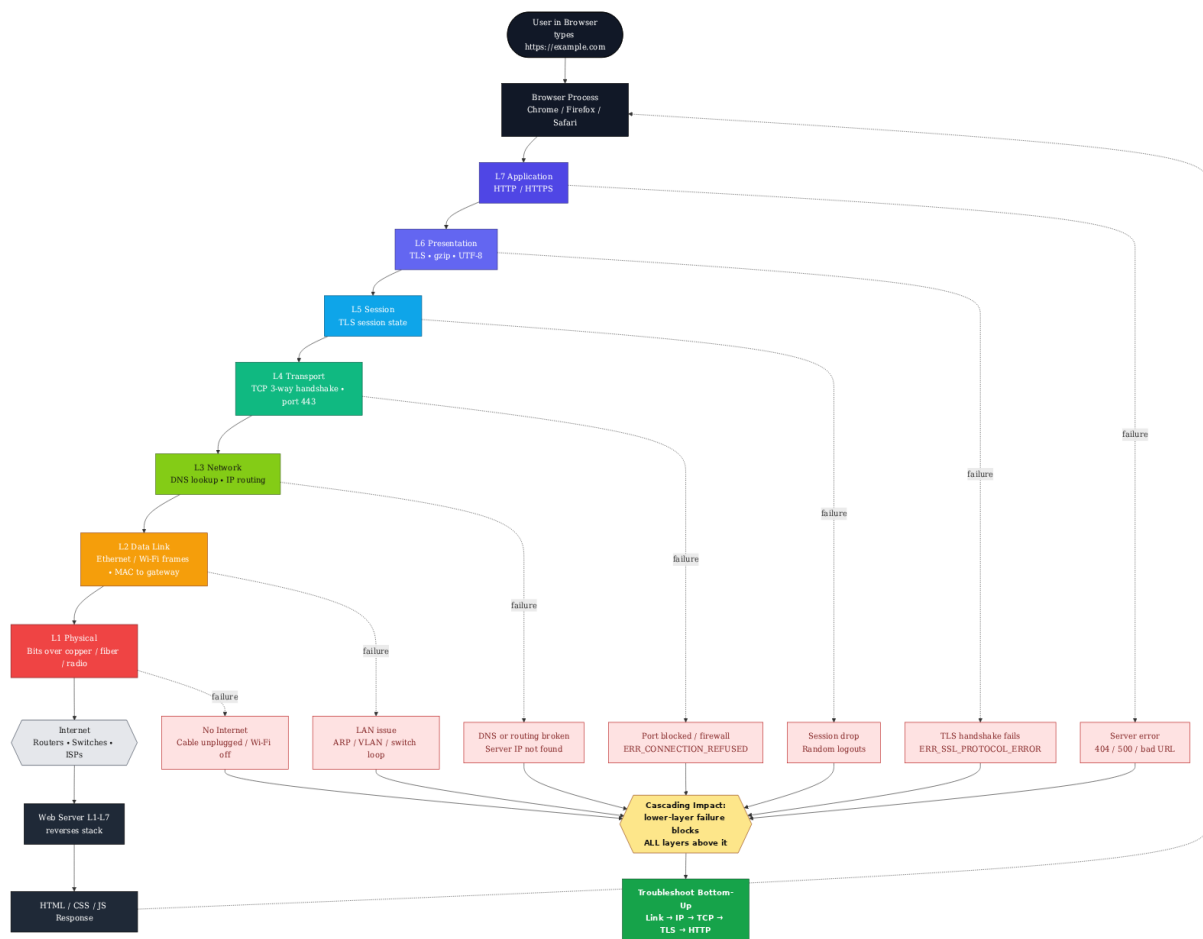


Figure 5 — Web browsing traced across all seven OSI layers.

Layer-by-Layer in a Web Request

Layer	Action while opening a webpage
Application (L7)	Browser issues HTTP/HTTPS request; DNS resolves URL to IP.
Presentation (L6)	TLS encryption / decryption and content encoding.
Session (L5)	Establishes and maintains the HTTP/TLS session.
Transport (L4)	TCP three-way handshake, segmentation, port 80/443.
Network (L3)	IP routing of packets through the internet.
Data Link (L2)	Framing and MAC delivery across each hop.
Physical (L1)	Signals over cable / fibre / Wi-Fi.

Effect of Failure at Each Layer

Failed Layer	Observed Symptom
Physical	No link light; browser: cannot connect / no network.
Data Link	MAC/switching issue; intermittent or no LAN connectivity.
Network	'Destination unreachable' / request times out — routing broken.
Transport	Connection refused / reset — TCP handshake fails.
Session	Session drops; TLS handshake or cookies fail.
Presentation	SSL/TLS certificate errors, garbled content.
Application	404 / 500 / DNS not found — protocol-level failure.

Because the layers are strictly hierarchical, failure in a lower layer blocks every layer above it, halting the entire communication.

Question 6 — Troubleshooting using a Concept Map

Given status: Physical ✓ Data Link ✓ Network ✗ Transport ✗ Application ✗

Concept Map

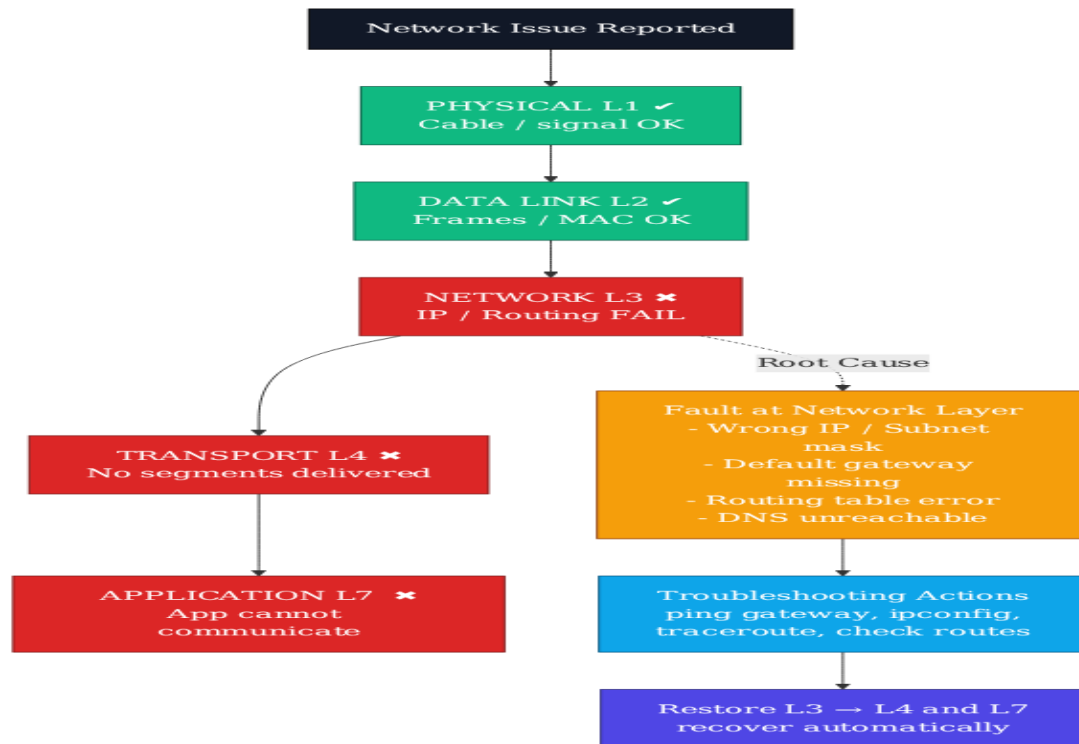


Figure 6 — Fault isolated to Network layer; upper layers fail as a consequence.

Analysis of the Problem

- Physical and Data Link are healthy → cable, NIC, switch and MAC-level frames all work.
- Network layer is the FIRST failing layer → this is the root cause.
- Transport and Application fail only because they depend on L3 to move packets between networks.

Likely Causes at Network Layer

- Incorrect IP address or subnet mask.
- Missing or wrong default gateway.
- Faulty routing table entry on a router.
- DNS server unreachable (name → IP resolution fails).

How the Concept Map Helps Troubleshoot

- Bottom-up scan quickly identifies the lowest failing layer — the true fault line.
- Focuses effort on L3 tools (ping gateway, ipconfig / ifconfig, traceroute, route table check) instead of guessing.

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RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation